PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal women: A systematic review and meta-analysis
AUTHORS	Kwak, Donghee; Ha, Jane; Won, Yousun; Kwon, Yeongkeun; Park, Sung-Soo

VERSION 1 – REVIEW

REVIEWER	Dr Poongkodi Karunakaran Government Mohan Kumaramangalam Medical College, Salem 636030, India
REVIEW RETURNED	17-Oct-2020

GENERAL COMMENTS	The authors have attempted systematic review and metanalysis of the effects of TSH suppression on bone mineral density (BMD) in post-menopausal woman undergoing thyroidectomy for thyroid cancer. The study is well-designed and easy to follow. However, the impact of varying level of TSH suppression on BMD was measured as standardized mean difference (SMD) of BMD between post-menopausal woman undergoing thyroidectomy for thyroid cancer and those of controls. The values of absolute BMD expressed in g/sq.cm was used to measure the outcome variable at the end of follow-up period. The authors have described that SMD was coded so that positive value represented higher BMD in the study subjects compared to controls with values of 0.2, .5 and 0.8 corresponding to small, moderate and larger effects. T-scores gives the number of SD of BMD from the mean of healthy 30-year old individual and Z-score is the number of standard deviations from the mean of age-, sex-, ethnicity-matched individual. Therefore, measurement of T-score and Z-score is the standard for the ordinal ranking of BMD and is therefore more appropriate for statistical analysis. Moreover, it removes confounding factors associated with age, sex and ethnicity. Intragroup analysis within the thyroidectomy group at different time points. i.e. the relative decline or increase in BMD from the baseline value at 5 years, 10 years of TSH suppression for varying levels will give more meaning data. Therefore, the authors should consider sub-group analyses. The authors have included total vs. near total thyroidectomy for sub-group analyses. However, the study by Franklyn which included subjects undergoing sub-total thyroidectomy could possibly bias the outcome. The term exogenous hyperthyroidism has to be changed as the term “hyperthyroidism” refers to the increased synthesis and release of thyroid hormone from the thyroid gland. Whereas, thyrotoxicosis denotes increased levels of thyroid hormone in the

	circulation resulting from either de-novo synthesis or exogenous administration. References should be written as superscript after punctuation mark through-out the manuscript. The following typo errors needs attention. Insert space between “thyroidectomydepending” Page 5, line 31. - The reviewer provided a marked copy with additional comments. Please contact the publisher for full details.

REVIEWER	Federico Hawkins Carranza Research Institute, Hospital 12 de Octubre, University Complutense of Madrid, Spain
REVIEW RETURNED	27-Oct-2020

GENERAL COMMENTS	The authors perform an interesting review and meta-analysis of an update discussion in the field of Clinical medicine, due to the long-term treatment with thyroxine hormones to suppress endogenous TSH, and the impact on bone metabolism. Introduction. -The authors express inline 8, that the effects of TSH suppression due to levothyroxine administration after thyroidectomy are unclear. They should clarify this because, in some reports, postmenopausal women lose bone after thyroidectomy due to thyroid carcinoma(Reverte et al, Endocr.Relat Ca.2005). -The authors express in paragraph 2, that no studies have investigated the effects of varying TSH levels on bone health. We recommend the authors to refer to the paper of Bauer (annals internal Med 2001)that describe that the risk of hip fractures is threefold increased in women with TSH suppressed below 0.1 mU/L, versus patients without this increase and TSH levels of 0.1-0. -We recommend the authors to refer to the previous meta-analysis, including the recent of Wang et al. Europ Review for Medical and Pharmacological Sciences 2020. Material and methods. The procedure is correct. The authors should refer is postsurgery hypoparathyroidism is included and not differentiated in the participants. Although reasons for the exclusion of 140 are summarized in supplemental Fig. 1, a brief commentary would be helpful to understand this screening. TSH levels reported in the studies were mean of various determination or a single time-point. The bone loss of the extracted data was made with the same technique. Were all done with DXA(Hologic or others) and the same for other variables/parameters as PTH. In table 1, the authors have include publications of patients with nonmedullary thyroid carcinoma. These patients were having other therapies. Effect of TSH suppression on the lumbar spine. The authors refer that overall, TSH suppression showed a detrimental effect on lumbar BMD. Was this the same for premenopausal and postmenopausal women ?. Detrimental effects were more pronounced concerning the extent of thyroidectomy: how many total or partial thyroidectomy was in the review. Discussion Is stated that “we identified stringent TSH suppression…. (means below0.1?). Did all the reviewed reports the number of patients under suppression below 0.1, so this can be said. The authors

	should comment on the ATA recommendations that high-risk patients should be controlled at <0.1, and also the recommendations for low-risk patients >0,5. On page 17 (second paragraph), the authors say_ In previous studies…….in postmenopausal women. Does this means that now this is not now observed….We observed a deleterious effect on the lumbar spine…. They should comment on the different response to thyroid hormones of trabecular (lumbar) vs. cortical (femoral neck)…This is described n page 18second paragraph. Should be once. The discussion should be shortened.

REVIEWER	Pardal-Refoyo , José Luis IBSAL, OTOLARYNGOLOGY
REVIEW RETURNED	19-Dec-2020

GENERAL COMMENTS	This is research through systematic review in PubMed, EMBASE and Cochrane Library, on bone pathology in patients with suppression of TSH following thyroid cancer thyroidectomy in postmenopausal women. Meta-analysis of 16 selected articles was performed analyzing the effects of TSH suppression measured in the femoral neck and lumbar spine in patients with thyroidectomy versus a control group. Material and methods The authors refer that the study was registered in PROSPERO under the number CRD42018104282. The PROSPERO database contains the registration entitled 'The effect of thyroidectomy on postoperative bone health: a systematic review and meta-analysis' and the authors indicate that they will include articles on 'Patients who received thyroidectomy due to any thyroid disease' with two versions (08 August 2018 and 24 January 2019). The selection of the type of patients does not match the registry. Authors must indicate this in the text. Results Page 9, line 20: discarded 141 articles (in the text indicate 140), in Figure 1 is the correct data Forest-plot figures are not quoted in the text and the figures are not numbered. Table 1 indicates that the number of postmenopausal female patients was 1417; the text indicates that it was 1127. Table 2 includes 22 studies, but it is not understood why the authors include it as it does not correspond to meta-analysis studies. I also don't understand the eligibility criterion. The authors discuss 'Effects of TSH suppression on the lumbar spine' and 'Effects of TSH suppression on the femoral neck'. Heterogeneity (I2) is very high. They also include subgroup analysis for both purposes. Other variables that may influence hypoparathyroidism or vitamin D levels are not evaluated. The authors indicate that to assess the risk of bias they 'adapted' three scales (references 21 to 23). Methodologically this is not correct.

REVIEWER	Awat Feizi

	Isfahan University of Medical Sciences, Department of Epidemiology and Biostatistics, School of Public Health, Isfahan University of Medical Sciences, Isfahan, Iran
REVIEW RETURNED	22-Jan-2021

GENERAL COMMENTS	Authors investigated the TSH suppression after thyroidectomy on BMD, through a systematic review and meta analysis - The title needs revision, as I see among many bone health variables , they investigated only BMD, then the title should be concentrated on BMD not a general term bone health, or they should investigate variety of bone health variables. - Abstract: some important databases such as ISI, SCOPUS, … have not been included. important point: it is not clear whether they evaluated the change before and TSH suppression after thyroidectomy and Levo therapy, it should be clarified both in the methods section of abstract as well as in presenting the study objective in introduction section. the presented results should be along with p-value, the results from "In the subgroup …." should be revised because they have not presented understandable. The suggestion for future studies should be on conducting prospective cohort studies on these patients, because they used only case-control studies. - First and second points regarding strengths and limitations do not make sound and reliable sense. - In literature review in introduction , please review the included studies in your systematic review and meta analysis briefly, it is necessary in the context of this study. - Methods section: important databases have not been searched, such as ISI, SCOPUS, .. Search strategy should be completed, how about the logical operators such as "OR" , AND" and combining these with relevant terms in the scope of this study? Please see published systematic review in good refereed journals for getting guide on how to organize this section, if you followed PRISMA it is expected you had done my suggested point for search! The literature search subsection is very defective, it is not clear which type of study designs have been included, although you said it as case-control in abstract, however it is needed to present this point in this section too. Some illegibility criteria such missing SD or SE are not restrictive condition, because you can extract it from range or IQR. How about the quality assessment? You missed this important and essential approach! Outcomes should be introduced more complete and clear, as stated earlier you should declare the outcomes have been collected at before and after thyroidectomy , I did not understand , because you did not clarify, if you have compared the cancer patient with whom? You should compare the study outcomes before and after thyroidectomy in cancer patients beside of comparing with controls, although you did not clarify what really control are. In data analysis you assumed comparable conditions between case and control that it is not seem to be sound assumption, you can consider some variables as confounders in meta regression and subgroup analyses. You also should evaluate the changes in BMDs in cancer patients group alone too beside of comparing it with controls. You should declare when you used fixed or random effects? please consider the relevant references for matters in statistical section such as for I-Squared values, and other ones. Those matters about publication bias needs major revisions, how about the subgroup analysis? how about the meta regression, how about the sensitivity analysis,

	all theses are important matters that unfortunately you did not followed them. Results: Please see results section of published meta-analyses in good journals, this section should be completed by considering my points mentioned above, you did not report p-values for the presented effect size, unfortunately you did not present any data about publication bias, meta regression, sensitivity analysis and some limited data on subgroup analysis, and some potential confounder and sources of heterogeneity for considering them in subgroup and meta regression analysis available you did not consider them. I strongly recommend to see published meta analyses in good refereed journals for getting guides on how to organize and complete your interesting study. Please research the databases for extending them till recent dates, I see till 2016!

VERSION 1 – AUTHOR RESPONSE

Referee #1

Comment #1

The authors have attempted systematic review and metanalysis of the effects of TSH suppression on bone mineral density (BMD) in post-menopausal woman undergoing thyroidectomy for thyroid cancer. The study is well-designed and easy to follow. However, the impact of varying level of TSH suppression on BMD was measured as standardized mean difference (SMD) of BMD between post-menopausal woman undergoing thyroidectomy for thyroid cancer and those of controls. The values of absolute BMD expressed in g/cm2 was used to measure the outcome variable at the end of follow-up period. The authors have described that SMD was coded so that positive value represented higher BMD in the study subjects compared to controls with values of 0.2, .5 and 0.8 corresponding to small, moderate and larger effects. T-scores gives the number of SD of BMD from the mean of healthy 30-year old individual and Z-score is the number of standard deviations from the mean of age-, sex-, ethnicity-matched individual. Therefore, measurement of T-score and Z-score is the standard for the ordinal ranking of BMD and is therefore more appropriate for statistical analysis. Moreover, it removes confounding factors associated with age, sex and ethnicity. Intragroup analysis within the thyroidectomy group at different time points. i.e. the relative decline or increase in BMD from the baseline value at 5 years, 10 years of TSH suppression for varying levels will give more meaning data. Therefore, the authors should consider sub-group analyses.

Response #1

We appreciate your comment. We agree that T-score and Z-score measurements are standard for the ordinal ranking of BMD and are more appropriate for the statistical analysis. However, few studies have measured T-scores and Z-scores. Therefore, we used BMD for the statistical analysis. Although the T-score and Z-score were not used in our statistical analysis, the study subjects were limited to postmenopausal women controlled for confounding factors associated with age and sex. At the time of the initial research design, we conducted a longitudinal analysis for the relative decline or increase in BMD from the baseline value of TSH suppression. However, there have been few studies, and the results were not significant. Further studies are needed to evaluate the longitudinal changes in BMD.

Comment #2

The authors have included total vs. near total thyroidectomy for sub-group analyses. However, the study by Franklyn which included subjects undergoing sub-total thyroidectomy could possibly bias the outcome.

Response #2

We agree with your comment. Franklyn's study, which included subjects that underwent subtotal thyroidectomy, was excluded from the total vs. near-total subgroup analysis. We have added more detail about this, as follows (Table 1): d This study which involved subjects that underwent subtotal thyroidectomy was excluded from the total versus near-total subgroup analysis.

Comment #3
The term exogenous hyperthyroidism has to be changed as the term "hyperthyroidism" refers to the increased synthesis and release of thyroid hormone from the thyroid gland. Whereas, thyrotoxicosis denotes increased levels of thyroid hormone in the circulation resulting from either de-novo synthesis or exogenous administration.
References should be written as superscript after punctuation mark through-out the manuscript. The following typo errors needs attention. Insert space between "thyroidectomydepending" Page 5, line 31.
Response #3
As per your suggestion, we have revised the term exogenous hyperthyroidism to "hyperthyroidism" (Page 22 Line 3). However, as per the BMJ formatting guidelines, we have formatted our references to be after the punctuation in square brackets. We have corrected the typing error "thyroidectomy depending" (Page 4, Line 15).

Referee #2
Comment #1
Introduction
The authors express inline 8, that the effects of TSH suppression due to levothyroxine administration after thyroidectomy are unclear. They should clarify this because, in some reports, postmenopausal women lose bone after thyroidectomy due to thyroid carcinoma (Reverte et al, Endocr.Relat Ca.2005).
Response #1
Thank you for your insightful comments. We have added more detail as follows (Page 4 Line 9): "However, the effects of TSH suppression due to levothyroxine administration after thyroidectomy are controversial.[10] Some studies reported a lack of deleterious effects on the bone mineral density (BMD) of long-term thyroxine suppressive therapy for differentiated thyroid carcinoma.[11-14] In contrast, other studies reported that thyroxine suppressive therapy had a detrimental effect on BMD.[15, 16]."

Comment #2
The authors express in paragraph 2, that no studies have investigated the effects of varying TSH levels on bone health. We recommend the authors to refer to the paper of Bauer (annals internal Med 2001) that describe that the risk of hip fractures is threefold increase in women with TSH suppressed below 0.1 mU/L, versus patients without this increase and TSH levels of 0.1-0.
Response #2
As per your suggestion, we have added more detail as follows (Page 4, Line 17): "One of them reported that females with a low TSH level (<0.1 mU/L) had a threefold increased risk for hip fractures and a fourfold increased risk for vertebral fractures compared with females who had normal TSH levels.[20] However, this study focused on the risk of fracture rather than BMD in females with low serum TSH levels."

Comment #3
We recommend the authors to refer to the previous meta-analysis, including the recent of Wang et al. Europ Review for Medical and Pharmacological Sciences 2020.
Response #3

We appreciate your insightful comment. We have added this information to the introduction. This study had a limitation because it did not perform a subgroup analysis according to the degree of TSH suppression, as in previous studies. We included more recent studies that were published in 2012. In addition, individual original studies included in this study were also examined and referenced to interpret the results.

Comment #4
The procedure is correct. The authors should refer is postsurgery hypoparathyroidism is included and not differentiated in the participants.
Although reasons for the exclusion of 140 are summarized in supplemental Fig. 1, a brief commentary would be helpful to understand this screening.
TSH levels reported in the studies were mean of various determination or a single time-point.
The bone loss of the extracted data was made with the same technique. Were all done with DXA(Hologic or others) and the same for other variables/parameters as PTH.
In table 1, the authors have include publications of patients with non-medullary thyroid carcinoma. These patients were having other therapies.
Effect of TSH suppression on the lumbar spine. The authors refer that overall, TSH suppression showed a detrimental effect on lumbar BMD. Was this the same for premenopausal and postmenopausal women? Detrimental effects were more pronounced concerning the extent of thyroidectomy: how many total or partial thyroidectomy was in the review.
Response #4
We appreciate your comment. We agree that this information has not been clearly presented. Thus, we modified the flow diagram of the literature selection process to show the reasons for the exclusion of 144 full-text articles. (Supplemental Figure 1).
As per your suggestion, we added more detail about the treatment of non-medullary thyroid carcinoma patients1, as follows (Table 1): c This study involved patients with non-medullary thyroid carcinoma without any kind of drug treatment known to influence bone turnover or diseases affecting calcium metabolism. Furthermore, we have added more detail about cancer etiology to our limitations, as follows (Page 26 Line 3): “Fifth, most of the cases were differentiated thyroid carcinoma, but thyroid cancer etiology was not reported, or patients with other etiologies were included. Therefore, additional treatments or differences in molecular pathogenesis may be the cause of heterogeneity.”
This study focused only on TSH suppression after thyroidectomy for thyroid cancer in postmenopausal women and did not present results for premenopausal women. However, due to our raw data analyses for premenopausal women, there was a protective effect of TSH suppression on the BMD of the lumbar spine and femoral neck. We have uploaded a figure of premenopausal women to the bottom of the paragraph as a supplemental material only for editors.
As shown in Figure 2A, there were eight studies with total thyroidectomy patients and five studies with near-total thyroidectomy patients.

Comment #5
Is stated that “we identified stringent TSH suppression…. (means below0.1?). Did all the reviewed reports the number of patients under suppression below 0.1, so this can be said. The authors should comment on the ATA recommendations that high-risk patients should be controlled at <0.1, and also the recommendations for low-risk patients >0,5.
On page 17 (second paragraph), the authors say_ In previous studies…….in postmenopausal women. Does this means that now this is not now observed….We observed a deleterious effect on the lumbar spine…. They should comment on the different response to thyroid hormones of trabecular (lumbar) vs. cortical (femoral neck)…This is described n page 18second paragraph. Should be once. The discussion should be shortened.
Response #5
Thank you for your insightful comments. It was not clarified whether all patients were high-risk or were suppressed below the TSH level of 0.1, so we divided them into study units based on the average

TSH level. This is mentioned in our limitations section, as follows (Page 26 Line 6): “Sixth, most studies did not clarify whether all of the included patients were high risk or suppressed below the TSH level of 0.1 mIU/L. Therefore, statistical analyses were performed based on the average TSH levels presented in each study.”
We agree with your recommendation. We already mentioned the ATA recommendations, as follows (Page 22 Line 15): “Clinical guidelines recommend the administration of levothyroxine as the optimal treatment for patients undergoing thyroidectomy for thyroid cancer. Patients with residual carcinoma or those at a high risk of cancer recurrence, those at an intermediate risk of cancer recurrence, and those at a low risk of cancer recurrence should maintain TSH levels at <0.1, 0.1–0.5, and 0.5–2 mIU/L, respectively.[1, 17, 18].” In addition, we have shortened the Discussion section as per your comments.

Referee #3
Comment #1
The authors refer that the study was registered in PROSPERO under the number CRD42018104282. The PROSPERO database contains the registration entitled ‘The effect of thyroidectomy on postoperative bone health: a systematic review and meta-analysis’ and the authors indicate that they will include articles on ‘Patients who received thyroidectomy due to any thyroid disease’ with two versions (08 August 2018 and 24 January 2019). The selection of the type of patients does not match the registry. Authors must indicate this in the text.
Response #1
Thank you for your insightful comments. As per your suggestion, we have revised the Materials and Methods section to detail why the patients selected do not match the PROSPERO registry. This is explained as (Page 6, Line 6): “At the time of initial registration of PROSPERO, patients who underwent thyroidectomy due to any thyroid disease were registered. However, there are few studies on BMD after thyroidectomy in typical hyperthyroidism diseases such as Graves’ disease, toxic multinodular goiter, and toxic adenoma, excluding cancer, so our study was limited to only patients with thyroid cancer.”

Comment #2
Page 9, line 20: discarded 141 articles (in the text indicate 140), in Figure 1 is the correct data
Forest-plot figures are not quoted in the text and the figures are not numbered.
Table 1 indicates that the number of postmenopausal female patients was 1417; the text indicates that it was 1127. Table 2 includes 22 studies, but it is not understood why the authors include it as it does not correspond to meta-analysis studies. I also don’t understand the eligibility criterion.

Response #2
We agree that this information was not clearly presented. We have revised the text to 141 articles to match Supplemental Figure 1. In addition, we have added a quotation to the forest plot and have modified Table 1 to match the text. A total of 1,127 postmenopausal female patients were included in the study. We also revised Table 2, which was inserted incorrectly.

Comment #3
The authors discuss ‘Effects of TSH suppression on the lumbar spine’ and ‘Effects of TSH suppression on the femoral neck’. Heterogeneity (I2) is very high. They also include subgroup analysis for both purposes.
Other variables that may influence hypoparathyroidism or vitamin D levels are not evaluated.
Response #3
We agree with your comments. As you mentioned, we observed high heterogeneity across the included studies for both effects of TSH suppression on the lumbar spine and femoral neck. Accordingly, we performed subgroup analyses (Figure 2), identified some possible sources of the heterogeneity, and partially resolved the heterogeneity.

We have added more detail to our limitation, as follows (Page 26 Line 8): “Seventh, other variables that may influence hypoparathyroidism or vitamin D levels, such as autoimmune conditions, genetic variants, radiation, mineral deposition, and magnesium deficiency, were not evaluated.[60]”

Comments #4
The authors indicate that to assess the risk of bias they 'adapted' three scales (references 21 to 23). Methodologically this is not correct.
Response #4
As per your suggestion, we have extensively revised our manuscript as follows (Page 8 Line 3): “The quality of the included studies was assessed using the Newcastle-Ottawa Quality Assessment Scale.[29] This assesses the quality of nonrandomized studies according to three variables: selection of the study population, comparability, and exposure factor. The quality scores are shown in Supplemental Table 3, with scores ranging from six to nine. All studies were found to be of moderate to high quality.” We also added Supplemental Table 3, which summarizes the quality assessment results using the Newcastle-Ottawa assessment scale.

Referee #4
Comment #1
The title needs revision, as I see among many bone health variables, they investigated only BMD, then the title should be concentrated on BMD not a general term bone health, or they should investigate variety of bone health variables.
Response #1
Thank you for your insightful comments. We agree with your suggestion. We have changed the title to “Effects of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal females: A systematic review and meta-analysis”.

Comment #2
Abstract: some important databases such as ISI, SCOPUS, … have not been included. Important point: it is not clear whether they evaluated the change before and TSH suppression after thyroidectomy and Levo therapy, it should be clarified both in the methods section of abstract as well as in presenting the study objective in introduction section.
The presented results should be along with p-value, the results from "In the subgroup …." should be revised because they have not presented understandable. The suggestion for future studies should be on conducting prospective cohort studies on these patients, because they used only case-control studies.
Response #2
Thank you for your insightful comments. We also searched the Scopus and Web of Science databases. The final search strategy and the results are presented in Supplementary Table 1. The p-values for the subgroup analyses are shown in Figure 2.
We agree that this information was not clearly presented. We have added more detail about our study selection to the abstract and clarified that only case control studies were included. Furthermore, we revised the results part of the abstract, as follows (Page 2 Line 19): “Subgroup analysis of the lumbar spine showed that the association between stringent TSH suppression and lower BMD was consistent among studies with >10 years of follow-up (SMD, -0.32; 95% CI, -0.50 to -0.14). Subgroup analysis of the femoral neck showed that total thyroidectomy was related to detrimental effects on the BMD of the femoral neck (SMD, -0.60; 95% CI, -0.89 to -0.31; I2 = 90.4%), but near-total thyroidectomy was not (SMD, 0.00; 95% CI, -0.30 to 0.30; I2 = 55.6%).”
We additionally revised our subgroup analysis results and added a description for the p-values (Page 20 Line 1-13).
We have added more detail to our limitation section, as follows (Page 26 Line 10): “Eighth, only case control studies were enrolled. A prospective cohort and longitudinal study are required.”

Comment #3
First and second points regarding strengths and limitations do not make sound and reliable sense. In literature review in introduction, please review the included studies in your systematic review and meta-analysis briefly, it is necessary in the context of this study.
Response #3
We agree with your comment. We have revised our strengths and limitations, as follows (Page 3 Line 12): “This is the first systematic review and meta-analysis to qualitatively collate and assess the results of studies that investigated the effect of thyroid stimulating hormone suppression after thyroidectomy for thyroid cancer on bone mineral density in postmenopausal females. We investigated the association between thyroid stimulating hormone suppression and lumbar spine or femoral neck bone mineral density in different subgroups (follow-up duration, surgical extent, ethnicity, and body mass index).”
In addition, we have added more detail about the review of the included studies in our systematic review and meta-analysis in the Introduction section, as follows (Page 4 Line 9): “However, the effects of TSH suppression due to levothyroxine administration after thyroidectomy are controversial.[10] Some studies reported a lack of deleterious effects on the bone mineral density (BMD) of long-term thyroxine suppressive therapy for differentiated thyroid carcinoma.[11-14] In contrast, other studies reported that thyroxine suppressive therapy had a detrimental effect on BMD.[15, 16]”

Comment #4
Search strategy should be completed, how about the logical operators such as "OR", AND" and combining these with relevant terms in the scope of this study? Please see published systematic review in good refereed journals for getting guide on how to organize this section, if you followed PRISMA it is expected you had done my suggested point for search! The literature search subsection is very defective, it is not clear which type of study designs have been included, although you said it as case-control in abstract, however it is needed to present this point in this section too. Some illegibility criteria such missing SD or SE are not restrictive condition, because you can extract it from range or IQR.
Response #4
We have presented the full search strategy in Supplemental Table 1. The logical operators were also utilized for a thorough literature search. We did not exclude studies with interquartile ranges. To make it clear, we modified the eligibility criteria as follows: (5) missing measures of “variability” (Page 7, Line 11).

Comment #5
How about the quality assessment? You missed this important and essential approach! Outcomes should be introduced more completely and clear, as stated earlier you should declare the outcomes have been collected at before and after thyroidectomy, I did not understand, because you did not clarify, if you have compared the cancer patient with whom? You should compare the study outcomes before and after thyroidectomy in cancer patients beside of comparing with controls, although you did not clarify what really control are.
In data analysis you assumed comparable conditions between case and control that it is not seem to be sound assumption, you can consider some variables as confounders in meta regression and subgroup analyses. You also should evaluate the changes in BMDs in cancer patients group alone too beside of comparing it with controls. You should declare when you used fixed or random effects? please consider the relevant references for matters in statistical section such as for I-Squared values, and other ones. Those matters about publication bias needs major revisions, how about the subgroup analysis? how about the meta regression, how about the sensitivity analysis, all these are important matters that unfortunately you did not follow them.
Response #5

We agree that this information was not clearly presented. We have added more detail about this, as follows (Page 7 Line 6): “control groups (matched healthy postmenopausal females were enrolled)”. All analyses were performed using random-effects models (DerSimonian and Laird method). Heterogeneity assessments were performed using the Mantel-Haenszel model, and publication bias assessments were performed using Begg’s and Egger’s tests. We have clearly stated the statistical methods in the “Data synthesis and analysis” section (Page 8).

Comment #6
Please see results section of published meta-analyses in good journals, this section should be completed by considering my points mentioned above, you did not report p-values for the presented effect size, unfortunately you did not present any data about publication bias, meta regression, sensitivity analysis and some limited data on subgroup analysis, and some potential confounder and sources of heterogeneity for considering them in subgroup and meta regression analysis available you did not consider them.
I strongly recommend to see published meta analyses in good refereed journals for getting guides on how to organize and complete your interesting study.
Please research the databases for extending them till recent dates, I see till 2016!
Response #6
Thank you for your insightful comments. We have added p-values for the presented effect size, shown in Figure 1 (Page 20 Line 8; Page 20 Line 10; Page 20 Line 20) and the p-value for the interaction test (Figure 2; Page 21 Line 7-8). We have added funnel plots that show no significant publication bias (Supplemental Figure 2; Page 20 Line 11-13; Page 20 Line 20-21).

REFERENCES

1. Giannini S, Nobile M, Sartori L, et al. Bone density and mineral metabolism in thyroidectomized patients treated with long-term L-thyroxine. Clin Sci 1994;87(5):593–7.

VERSION 2 – REVIEW

REVIEWER	Karunakaran, Poongkodi Government Mohan Kumaramangalam Medical College, Endocrine surgery
REVIEW RETURNED	25-Apr-2021

GENERAL COMMENTS	The authors have addressed the issue on "The effect of TSH suppression after thyroidectomy for thyroid cancer on BMD in post-menopausal women". A systematic review and meta-analysis which is first of its kind and important in the present scenario which is seeing a rise in thyroid cancer incidence. The following issues have to be addressed by the authors: 1. 'Men and women' is more appropriate terminology than males and females. 2. Reference numbering shall be written as superscript after the punctuation mark. 3. Syntax and grammatical errors to be addressed. 4. supplemental Figure 1. Flow-diagram in page 45 of 89. Lines 27-28. The number adds to 153 and not 144. Kindly check. - The reviewer provided a marked copy with additional comments. Please contact the publisher for full details.

VERSION 2 – AUTHOR RESPONSE

Referee #1

Comment #1

'Men and women' is more appropriate terminology than males and females.

Response #1

Thanks for your comment. We have changed the terminology per your suggestion.

Comment #2

Reference numbering shall be written as superscript after the punctuation mark.

Response #2

We have changed the reference numbering per your suggestion.

Comment #3

Syntax and grammatical errors to be addressed.

Response #3

We appreciate your kind comments and editing. We made changes per your suggestions, and then two authors thoroughly proofread the manuscript and addressed some grammatical errors.

Comment #4

Supplemental Figure 1. Flow-diagram in page 45 of 89. Lines 27-28. The number adds to 153 and not 144. Kindly check.

Response #4

Thank you for your insightful comment. We recognized the error and fixed it.